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TASK MANAGEMENT SYSTEM

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Abstract

Every project or event involves a series of tasks, activities, people, budget and deadlines. No matter how big or small the project is, the success or failure of the project depends on the execution plan and the order in which the tasks are fulfilled. It takes a great deal of skill to do this well, the time invested in building good project management techniques can pay off enormously and helps to achieve projects on time within the resource constraints. When one knows how to organize, schedule and delegate tasks it is an easy step for them to obtain the success of the project.

Task management is the process of managing a task through its lifecycle. It involves planning, testing, tracking, and reporting. Task management can help either individuals achieve goals, or groups of individuals collaborate and share knowledge for the accomplishment of collective goals. Tasks are also differentiated by complexity, from low to high.

Currently, there are many tools of Task Management Systems such as To-Do, Microsoft Planner, Kanban Boards, Microsoft Project and others. However mostly Task Management are built build in another tool, reserve our Python code as you may know that Python is very popular, and fast performance compared to other languages, so aim to develop this Task Management System to help user able to manage their task easily as well as backup developer easy for coding.

Keywords: paper-based, Task Management System (TMS), Tkinter, Python, SQLite

1. Introduction

A task management system (TMS) is a tool or process that helps individuals and teams organize, prioritize, and track their work. TMS can be used to manage tasks of all sizes, from simple to-do items to complex projects.

TMS typically include features such as:

- Task creation and editing
- Due dates and priorities
- Task assignment and delegation
- Subtasks and dependencies
- File attachments
- Comments and collaboration
- Reporting and analytics

TMS can be used in a variety of settings, including:

- Personal productivity
- Teamwork and collaboration
- Project management
- Business process management

There are many benefits to using a task management system, including:

- Increased productivity
- Improved focus
- Reduced stress
- Better teamwork and collaboration
- Enhanced project visibility and control
- Improved time management
- Reduced costs
- Increased profits

2. Literature Review

Task management is a critical aspect of personal and professional productivity. With the increasing complexity of modern life and work, efficient task management systems have become indispensable. This literature review explores the key components and principles of task management systems, their impact on productivity, and the evolving landscape of digital task management tools.

Early task management systems were primarily paper based, using to-do lists and planners. However, with the advent of personal computers and digital technology, task management tools evolved. The seminal work of **David Allen**, as outlined in "Getting Things Done," revolutionized the way individuals approached task management by introducing a methodology that emphasizes clear organization, prioritization, and action.

Features list task management system:

- Task creation and editing: Users can create and edit tasks, including setting due dates, priorities, and descriptions.
- Task assignment and delegation: Users can assign and delegate tasks to other individuals or teams.
- File attachments: Users can attach files to tasks, such as project documents, images, or videos.
- Comments and collaboration: Users can leave comments and collaborate on tasks with other team members.
- Reporting and analytics: TMS typically includes reporting and analytics features that can help users to track their progress, identify bottlenecks, and make improvements.

In addition to these core features, some TMS also offer additional features such as:

- Time tracking: Users can track the time they spend on tasks, which can help to improve time management and productivity.
- Calendars and scheduling: Users can create and manage calendars and schedules, which can help to coordinate tasks and deadlines.
- Kanban boards and Gantt charts: Users can create and use Kanban boards and Gantt charts to visualize their work and track progress.
- Integrations: TMS can be integrated with other software tools and platforms, such as project management tools, CRM systems, and email clients.

The specific features that are included in a TMS will vary depending on the specific needs of the individual or team. However, the core features listed above are common to most TMS.

3. Material and Methods

3.1. Material

3.1.1. Python-TKinter



The programming language Python was conceived in the late 1980s, and its implementation was started in **December 1989 by Guido van Rossum** at **CWI in the Netherlands** as a successor to ABC capable of exception handling and interfacing with the Amoeba operating system. Van Rossum is Python's principal author, and his continuing central role in deciding the direction of Python is reflected in the title given to him by the Python community, Benevolent Dictator for Life (BDFL). However, Van Rossum stepped down as leader on July 12, 2018. Python was named after the BBC TV show Monty Python's Flying Circus.

Tikinter is the older name for the Tkinter library in Python. Tkinter is a Python interface to the Tk GUI toolkit, which is a cross-platform GUI toolkit that is included in the standard Python distribution. Tkinter is used to create graphical user interfaces (GUIs) for Python applications.

The name "**Tkinter**" was used for the Tkinter library in Python 2.x. In Python 3.x, the library was renamed to "Tkinter". This was done to simplify the name of the library and to make it more consistent with the other Python libraries

Python's evolution has been characterized by a commitment to simplicity, readability, and a strong emphasis on code maintainability. This evolutionary trajectory has culminated in Python 3, a robust and modern iteration of the language.

Python is a general-purpose, high-level programming language. It is known for its simple, easy-to-read syntax and its large standard library. Python is used in a wide variety of applications, including web development, data science, machine learning, and artificial intelligence.

Some of the key features of Python include:

- Simple and easy-to-read syntax: Python code is written in a simple and easy-to-read syntax, which makes it a good language for beginners to learn.
- Large standard library: Python comes with a large standard library, which includes a variety of modules and functions that can be used for common tasks such as file I/O, networking, and data manipulation.
- Multi-paradigm programming: Python supports multiple programming paradigms, including object-oriented programming, procedural programming, and functional programming. This makes it a versatile language that can be used for a variety of different types of applications.
- Dynamic typing: Python is a dynamically typed language, which means that the type of a variable is not declared explicitly. This makes Python code more flexible and easier to write.
- Garbage collection: Python has automatic garbage collection, which means that the programmer does not need to worry about explicitly freeing memory. This makes Python code more concise and easier to maintain.

Python is a popular language for both beginners and experienced programmers. It is a good choice for a variety of different types of applications, and it is relatively easy to learn.

Here are some examples of applications that are built using Python:

- Web frameworks: Django, Flask, Pyramid
- Data science libraries: NumPy, Pandas, SciPy
- Machine learning libraries: TensorFlow, PyTorch, Scikit-learn
- Artificial intelligence libraries: OpenAI Gym, TensorFlow Agents
- Scripting languages: Ansible, SaltStack, Puppet
- Desktop applications: PyQt, wxPython, Kivy

3.1.2. SQLite



In our Task Management System Assignment, we use SQLite as our database system because it is an embedded database, SQLite allows developers to store data without needing a separate database server. It is helpful for small data needs, such as mobile devices and other applications that require low-memory footprint systems.

D. Richard Hipp designed SQLite in the spring of 2000 while working for General Dynamics on contract with the United States Navy. Hipp was designing software used for a damage-control system aboard guided-missile destroyers, which originally used HP-UX with an IBM Informix database back end. SQLite began as a Tcl extension.

In August 2000, version 1.0 of SQLite was released, with storage based on gdbm (GNU Database Manager). In September 2001, SQLite 2.0 replaced gdbm with a custom B-tree implementation, adding transaction capability. In June 2004, SQLite 3.0 added internationalization, manifest typing, and other major improvements, partially funded by America Online. In 2011, Hipp announced his plans to add a NoSQL interface to SQLite, as well as announcing UnQL, a functional superset of SQL designed for document-oriented databases. In 2018, SQLite adopted a Code of Conduct based on the Rule of Saint Benedict which caused some controversy and was later renamed as a Code of Ethics.

SQLite is a lightweight database management system (DBMS) that stores data in a single file. It is open-source and public domain software, meaning that it is free to use and distribute. SQLite is known for its portability, efficiency, and reliability.

SQLite is a popular choice for embedded systems and mobile applications because of its small footprint and low resource requirements. It is also used in many desktop and server applications, such as web browsers, email clients, and word processors.

SQLite supports a variety of SQL features, including:

- Tables and views
- Indexes and constraints
- Transactions and rollbacks
- Subqueries and joins

SQLite is a good choice for a variety of applications, including:

- Mobile applications
- Embedded systems
- Desktop applications
- Server applications
- Web applications

Here are some examples of popular applications that use SQLite:

- Firefox
- Chrome
- Opera
- Thunderbird
- Spotify
- VLC Media Player
- WordPress
- Drupal
- Joomla

SQLite is a powerful and versatile database management system that is used in a wide variety of applications. It is a good choice for both beginners and experienced database users

3.1.3. GitHub



In our Task Management System, we are using GitHub to store our code.

GitHub is a web-based hosting service for version control using Git. It is used to store code, track changes, collaborate on projects, and build software.

GitHub offers a variety of features to help developers work together effectively, including:

- Code hosting: GitHub provides a place to store your code and track changes over time.
- Issue tracking: GitHub allows you to track bugs and feature requests in your projects.
- Pull requests: GitHub allows you to submit changes to your code for review by other developers.
- Collaboration tools: GitHub provides tools to help you collaborate with other developers on your projects, such as wikis, discussions, and code review.
- Deployment tools: GitHub provides tools to help you deploy your software to production.

GitHub is used by a wide variety of organizations, from small startups to large enterprises. It is a popular choice for open-source projects, as it provides a way for developers from all over the world to collaborate on code.

Here are some of the benefits of using GitHub:

- Code version control: GitHub allows you to track changes to your code over time, which can help you to recover from mistakes and revert to previous versions of your code.
- Collaboration: GitHub makes it easy to collaborate with other developers on your projects. You can share your code with others, review their changes, and discuss your project.
- Deployment: GitHub can help you to deploy your software to production. You can use GitHub's deployment tools to automate the process of deploying your code to a server.

- Community: GitHub has a large and active community of developers. You can ask questions, get help, and share your code with others.

Overall, GitHub is a powerful and versatile platform that can help you to manage your code, collaborate with others, and build software. It is a good choice for both beginners and experienced developers.

3.2. Methods

3.2.1. Admin Application

For the current version of application, we only pre-define users in our Task Management System, meaning that admin application must register user via our backend developer team. After users create then the user will be able to login to perform the operation.

However, our further discussion will develop more function as below:

- Admin User will be able to approval for the user request register
- Admin User have right to manage the task, create, Update, Delete and modify each task if needed.
- Admin will be able to see all tasks in each user

Below our current coding from admin backend design.

```
login.py
1 from tkinter import *
2 from db.db import Database
3
4 db = Database("../db/TASK_MANAGEMENT.db")
5
6 def handleLogin():
7     user = username.get()
8     pwd = password.get()
9     responseMessage = db.login(user, pwd)
10    print(responseMessage)
11    loginMessage = responseMessage['message']
12
13    loginEmployeeId = responseMessage['employeeId'] if 'employeeId' in responseMessage else ''
14    message.set(loginMessage)
15
16    # on success close login form
17    if loginEmployeeId:
18        login_screen.destroy()
19        import index
20        index.homePage(responseMessage, c
21
22 def Logininform():
23     global login_screen
24     login_screen = Tk()
25     #Setting title of screen
26     login_screen.title("Login | Task Man
27     #setting height and width of screen
28     login_screen.geometry("350x250")
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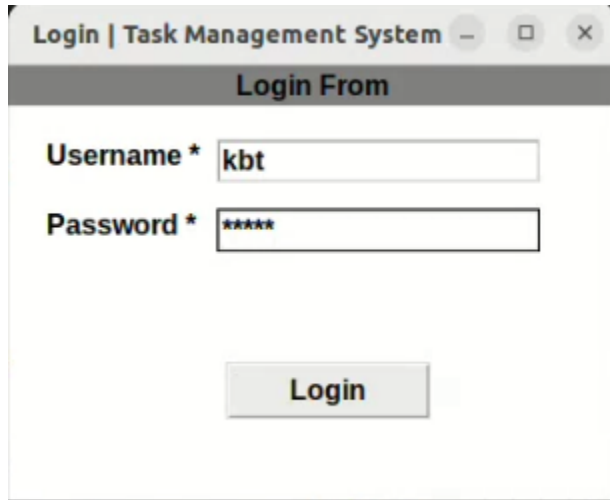
/bin/python3.10 /home/kbt/Documents/personal/UP/Python/task-management/login.py
task-management git:(main) x /bin/python3.10 /home/kbt/Documents/personal/UP/Python/task-management/login.py
(4, 'Thai', 'thai', '12345')
{'employeeId': 4, 'id': 4, 'name': 'Thai', 'message': 'Login success'}
task-management git:(main) x /bin/python3.10 /home/kbt/Documents/personal/UP/Python/task-management/login.py

```

3.2.2. User Application

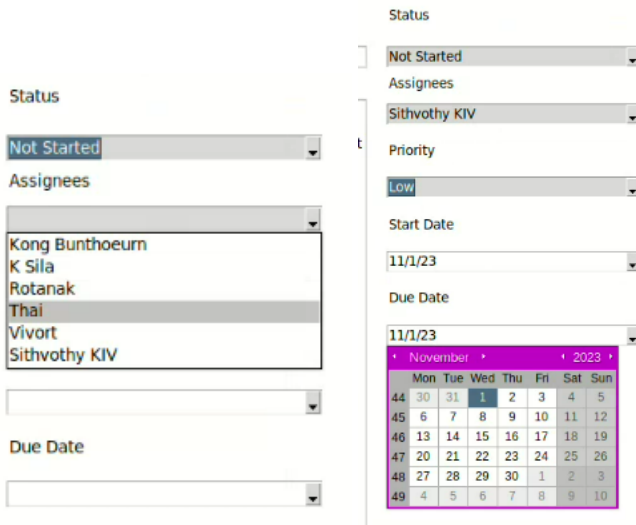
For the current version of application, we only pre-define users in our Task Management System, meaning that we must register user via our backend developer team. After users create then the user will be able to login to perform the operation.

when they execute application, it will appear login screen.



And user required to input **Username** and **Password**

As we have mentioned that we have pre-define user in backend system so we can only select users in drop down list, and other items as below:



4. Result

4.1. Workflow

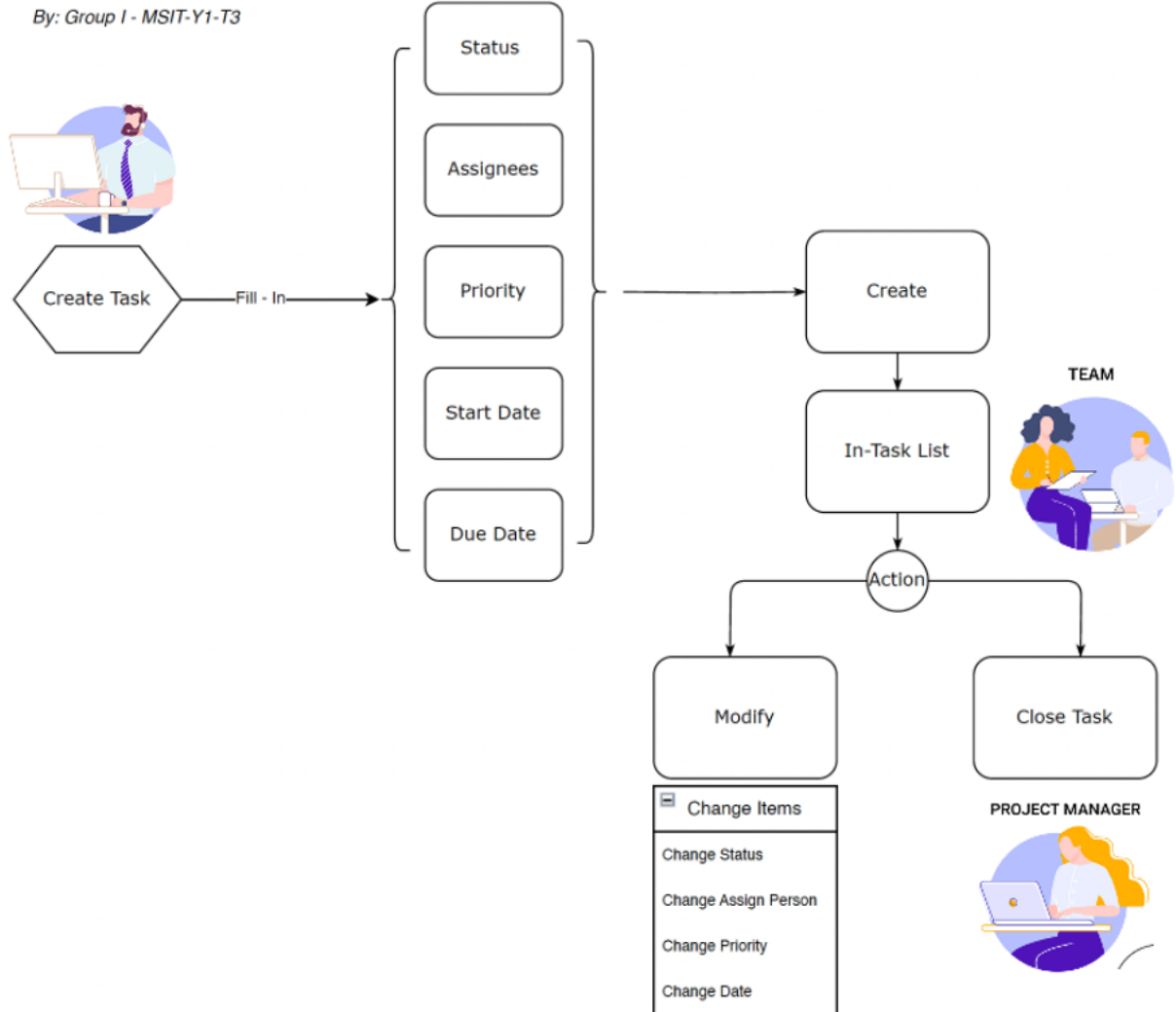
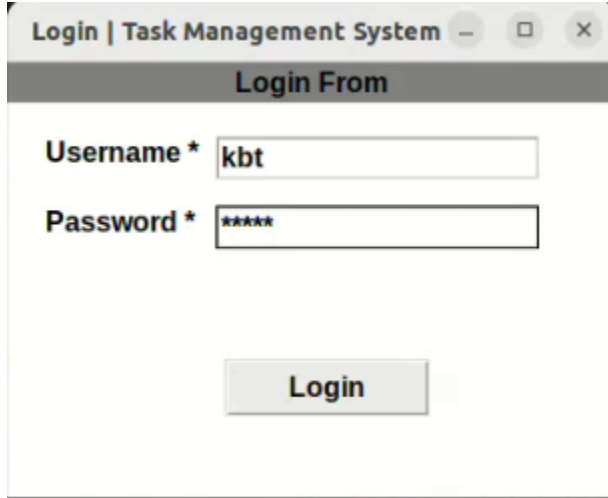


Figure #1. The workflow of Task Management System our first version from Python.

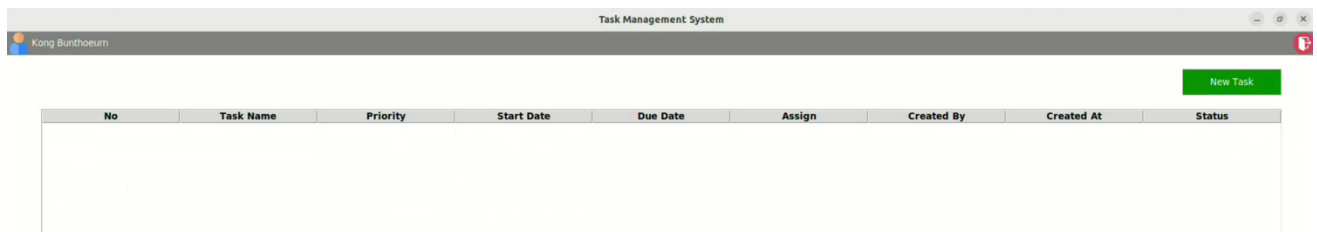
4.2. User's Guide

When application start, we require user to **login** with below login Window form



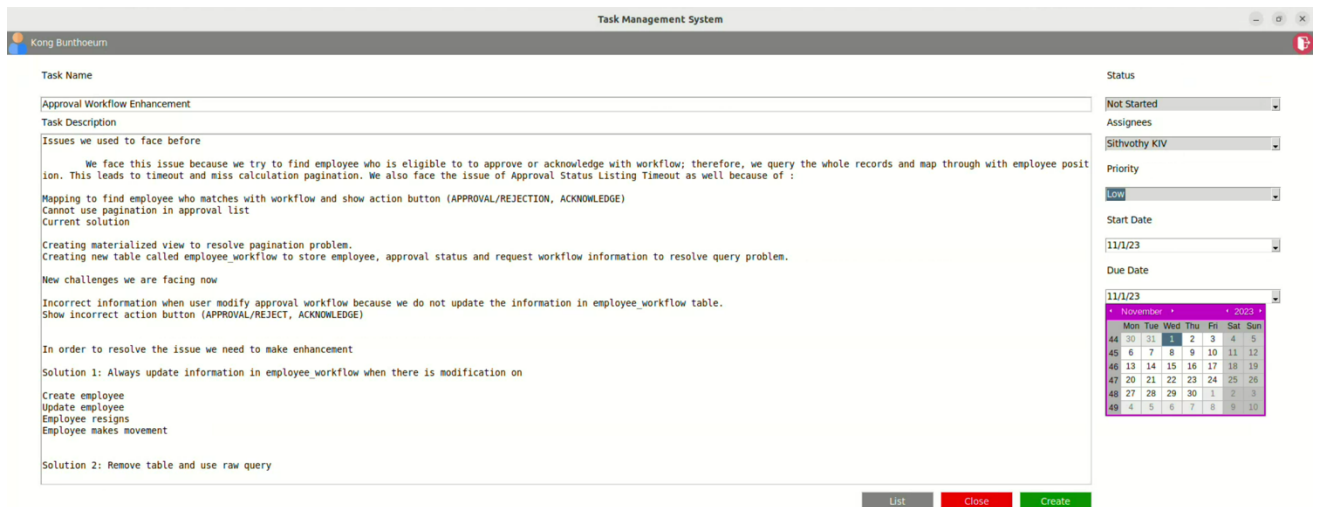
The login window is titled "Login | Task Management System". It has a "Login From" header. Below it, there are two input fields: "Username *" with the value "kbt" and "Password *" with masked characters "*****". A "Login" button is positioned below the password field.

User can click on **New Task** to Create New Task



The dashboard shows a table with columns: No, Task Name, Priority, Start Date, Due Date, Assign, Created By, Created At, and Status. A "New Task" button is located in the top right corner.

Then they can Input some Fill such as; **Task Name**, **Task Description**, **Status**, **Assignees**, **Priority**, **Start** and **Due Date**, then click on **Create**.

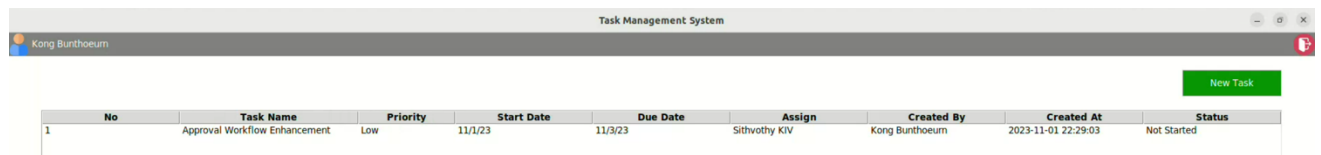


The task creation form includes the following fields:

- Task Name:** Approval Workflow Enhancement
- Task Description:** Issues we used to face before
We face this issue because we try to find employee who is eligible to approve or acknowledge with workflow; therefore, we query the whole records and map through with employee position. This leads to timeout and miss calculation pagination. We also face the issue of Approval Status Listing Timeout as well because of :
Mapping to find employee who matches with workflow and show action button (APPROVAL/REJECTION, ACKNOWLEDGE)
Cannot use pagination in approval list
Current solution
Creating materialized view to resolve pagination problem.
Creating new table called employee_workflow to store employee, approval status and request workflow information to resolve query problem.
New challenges we are facing now
Incorrect information when user modify approval workflow because we do not update the information in employee_workflow table.
Show incorrect action button (APPROVAL/REJECT, ACKNOWLEDGE)
In order to resolve the issue we need to make enhancement
Solution 1: Always update information in employee_workflow when there is modification on
Create employee
Update employee
Employee resigns
Employee makes movement
Solution 2: Remove table and use raw query
- Status:** Not Started
- Assignees:** Sithvothy KIV
- Priority:** Low
- Start Date:** 11/1/23
- Due Date:** 11/1/23

Buttons at the bottom: List, Close, Create.

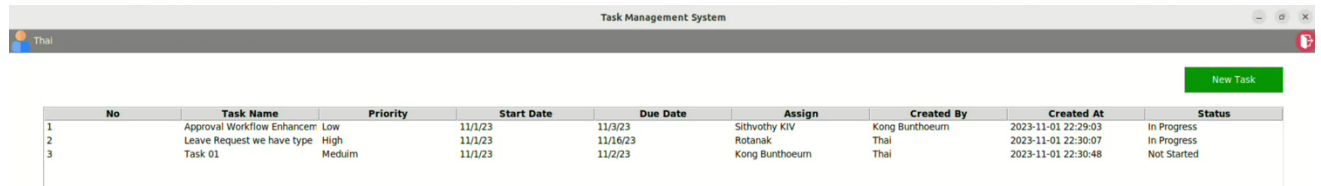
After click Create, you will see your result of Task as below:



The screenshot shows a web browser window titled "Task Management System". The user profile "Kong Burthoeum" is visible in the top left. A green "New Task" button is in the top right. Below is a table with one task entry.

No	Task Name	Priority	Start Date	Due Date	Assign	Created By	Created At	Status
1	Approval Workflow Enhancement	Low	11/1/23	11/3/23	Sithvothy KIV	Kong Burthoeum	2023-11-01 22:29:03	Not Started

You can each *Priority*, *Low*, *High*, *Medium*, also assign to who? And status of tasks in the main screen after user login as below:



The screenshot shows the same "Task Management System" interface, but now with three tasks listed in the table.

No	Task Name	Priority	Start Date	Due Date	Assign	Created By	Created At	Status
1	Approval Workflow Enhancement	Low	11/1/23	11/3/23	Sithvothy KIV	Kong Burthoeum	2023-11-01 22:29:03	In Progress
2	Leave Request we have type	High	11/1/23	11/16/23	Rotanak	Thai	2023-11-01 22:30:07	In Progress
3	Task 01	Medium	11/1/23	11/2/23	Kong Burthoeum	Thai	2023-11-01 22:30:48	Not Started

When choosing a task management system, it is important to consider the specific needs of the individual or team. Some factors to consider include:

- The size and complexity of the tasks to be managed
- The need for collaboration features
- The need for reporting and analytics
- The budget
- The platform (cloud-based, on-premises, or hybrid)

Using a task management system effectively

Once a task management system has been chosen, it is important to use it effectively. Some tips for using a TMS effectively include:

- Clearly define the tasks to be managed
- Prioritize tasks and set deadlines
- Delegate tasks to appropriate individuals or teams
- Track progress and adjust as needed
- Communicate with others about task assignments and deadlines
- Use reporting and analytics to identify areas for improvement

5. Discussion and Future Work

Our result of this Task Management System version 1.0 by our Python and SQLite was very limited, however we still can use it as basic Task Creation, and assignment to team, put priority also include schedule plan.

We have very short of times to develop it, also our knowledge of Python language still not yet advance. We do need to learn more on Python also database in order to accomplish a very standard application function as well as better UI.

For the future work in task management should not only focus on technological advancements but also take into account human factors, user experience, and the changing dynamics of work in the digital age. Research and development efforts should aim to simplify and enhance the way individuals and organizations manage tasks to promote greater efficiency, productivity, and well-being.

Here are some of the areas where future research on task management systems could focus:

- Developing new AI and ML algorithms to improve the performance and accuracy of TMS.
- Exploring new ways to integrate TMS with other software tools and platforms.
- Developing new context-aware features for TMS.
- Investigating the impact of TMS on individual and team performance.
- Developing new training and support resources for users of TM

Here are some of the key trends in task management systems:

- Artificial intelligence (AI) and machine learning (ML): AI and ML are being used to develop new features and functionality for TMS. For example, AI can be used to automatically prioritize tasks, identify bottlenecks, and provide recommendations to users.
- Integration with other software tools and platforms: TMS are increasingly being integrated with other software tools and platforms, such as project management tools, CRM systems, and email clients. This integration allows users to manage their tasks from within a single interface and to automate workflows.
- Context-awareness: TMS are becoming more context-aware, meaning that they can adapt to the user's needs and environment. For example, a context aware TMS might automatically suggest tasks to the user based on their current location or the time of day.

6. Conclusion

In conclusion, our Task Management System we are still in developing state as we have mentioned in "Discussion and Future Work" session. However, our application is still useful for better manage Tasks instead of traditional paper based.

The results in this Task Management System research and development article present a methodology in scientific research in computer software development to guide students and computer science students to be competent in capacity development. Or coding skills in Python and understand how to write a professional study essay.

And to help workers be able to divide work easily and manage the work situation better. This is just a small part of helping employees better organize and manage their work.

Task management systems have evolved significantly, from traditional paper-based methods to sophisticated digital tools. The research indicates that effective task management positively influences productivity and work-life balance. As technology continues to advance, the task management landscape will undergo further changes, offering new opportunities and challenges for individuals and organizations.

This literature review gives an overview of the historical development, key components, impact, and challenges of task management systems, and insights into the evolving role of digital tools in this context. Further research in this field is essential to better understand how individuals and organizations can maximize the benefits of effective task management.

For the future, students and other researchers can write this task management program to be better and more up to date, can create more Interface or Function can be used on both IOS and Android phones to increase convenience Another level.

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